



Barker
College

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Student Number

2023

**YEAR 12 ASSESSMENT BLOCK
SEMESTER 1**

Mathematics Extension 2

Section I - Multiple Choice

Write your student number at the top of this page.

Remove this sheet.

Select the alternative A, B C or D that best answers the question. Fill in the response oval completely.

Sample $2 + 4 =$ A 2 B 6 C 8 D 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
 correct

**Start
Here** →

1. A ☐ B ☐ C ☐ D ☐

5. A ☐ B ☐ C ☐ D ☐

2. A ☐ B ☐ C ☐ D ☐

6. A ☐ B ☐ C ☐ D ☐

3. A ☐ B ☐ C ☐ D ☐

7. A ☐ B ☐ C ☐ D ☐

4. A ☐ B ☐ C ☐ D ☐

8. A ☐ B ☐ C ☐ D ☐

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Student Number

2023

**YEAR 12
ASSESSMENT BLOCK**

Mathematics Extension 2

Staff Involved:

BHC (Mr Carruthers)
DZP (Mr Peattie)
ARM* (Mr Mallam)

**12:00 NOON
FRIDAY 31 MARCH**

35 copies

TOPICS COVERED:

Complex Numbers I & II

Proof

General Instructions

- Working time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided separately
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- Marks may not be awarded for careless or poorly arranged working
- Diagrams are not to scale
- Write your student number at the top of this page

**Total marks:
80**

Section I – 8 marks (pages 4-6)

- Attempt Questions 1-8
- Allow about 15 minutes for this section

Section II – 72 marks (pages 7-12)

- Attempt Questions 9-14
- Allow about 1 hour and 45 minutes for this section

Section I

8 marks

Attempt Questions 1-8

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1-8.

1 What is the negation of the statement: “If it is raining, then I carry an umbrella”?

- A. If it is not raining, then I do not carry an umbrella.
- B. If I am carrying an umbrella, then it is raining.
- C. It is not raining and I am carrying an umbrella.
- D. It is raining and I am not carrying an umbrella.

2 What is the value of $(4 - 3i)^2$?

- A. $7 - 24i$
- B. $7 + 24i$
- C. $25 - 24i$
- D. $25 + 24i$

3 If n is a positive integer, for how many values of n is $9n^2 - 4$ a prime number?

- A. 0
- B. 1
- C. 2
- D. More than 2

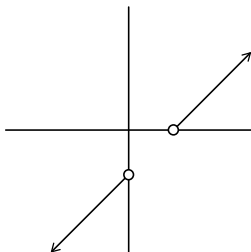
- 4 Which of the following best describes the locus of points in the Argand plane given by

$$|z - 2| = 2|z + 1| ?$$

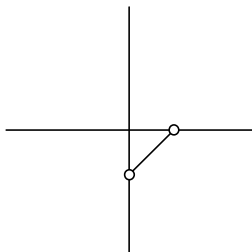
- A. A vertical line through (0, 0).
- B. A vertical line through (1, 0).
- C. A circle with centre (0, 0).
- D. A circle with centre (-2, 0).

- 5 Which diagram best represents the solutions to the equation $\text{Arg}(z - 2) = \text{Arg}(z + 2i)$?

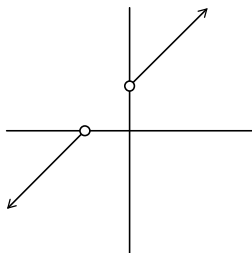
A.



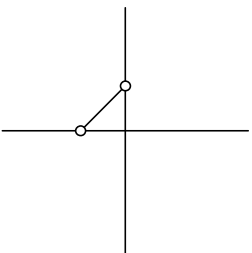
B.



C.



D.



- 6 For complex z , if $\text{Im}(z) > 0$, what is the value of $\text{Arg}\left(\frac{z\bar{z}}{z - \bar{z}}\right)$?

A. $-\frac{\pi}{2}$

B. 0

C. $\frac{\pi}{2}$

D. π

- 7 For integers x and y , which one of the following statements is equivalent to the statement: "There is no integer which is the largest integer"?

A. $\exists y (\forall x, y > x)$

B. $\forall x (\exists y, y > x)$

C. $\exists x (\forall y, y > x)$

D. $\forall y (\exists x, y > x)$

- 8 Let $(x + iy)^{18} = a + ib$ for $a, b, x, y \in \mathbb{R}$. Which expression is equivalent to $(y - ix)^{18}$?

A. $-a - ib$

B. $b - ia$

C. $-b + ia$

D. $-a + ib$

End of Section I

Section II

72 marks

Attempt Questions 9–14

Allow about 1 hour and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

Question 9 (12 marks) Use a SEPARATE writing booklet.

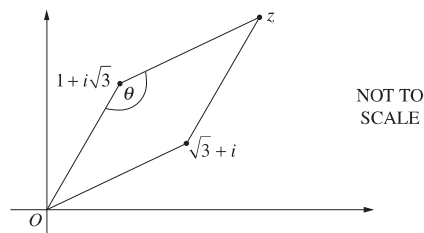
(a) Consider the complex numbers $w = 2 - 3i$ and $z = 3 + 4i$.

(i) Find $\bar{w} + z$. 1

(ii) Find $|w|$. 1

(iii) Express $\frac{w}{z}$ in the simplest form of $x + iy$ where x and y are real. 2

(b) On the Argand diagram below, the complex numbers 0 , $1 + i\sqrt{3}$, $\sqrt{3} + i$ and z form a rhombus.



(i) Find z in the form $x + iy$. 1

(ii) An interior angle of the rhombus, θ , is marked on the diagram. Find the value of θ . 2

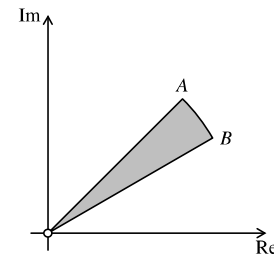
(c) Use proof by contradiction to show that $\log_2 7$ is irrational. 2

(d) Prove that the sum of two consecutive positive even powers of 3 is always a multiple of 90. 3

Question 10 (12 marks) Use a SEPARATE writing booklet.

(a) The diagram below shows the region on the complex plane given by:

$$|z| \leq 4\sqrt{2} \text{ and } \frac{\pi}{6} \leq \text{Arg}(z) \leq \frac{\pi}{4}.$$



(i) Find the complex number represented by point A in modulus-argument form. 1

(ii) Find the complex number represented by point B in the form $x + iy$. 1

(iii) Determine whether $5 + 2i$ lies within the region. 2

(b) (i) Given that $\cos \theta + i \sin \theta = e^{i\theta}$, prove de Moivre's theorem: 1
for positive integers n , $(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta)$.

(ii) Using de Moivre's theorem, or otherwise, show that, for positive integers n , 2

$$(1 + i)^n + (1 - i)^n = 2 \left(\sqrt{2} \right)^n \cos \frac{n\pi}{4}.$$

(c) Two of the zeros of the polynomial $P(x) = x^4 - 4x^3 + 11x^2 - 14x + 10$ are $a + ib$ and $a + 2ib$, where a and b are real and $b > 0$.

(i) Find the values of a and b . 3

(ii) Hence, or otherwise, express $P(x)$ as a product of quadratic factors with real coefficients. 2

Question 11 (12 marks) Use a SEPARATE writing booklet.

- (a) Consider the following statement for integers m and n .

P : If $\frac{mn}{2}$ is an integer then m or n must be even.

- (i) Write down the contrapositive of P . 1
- (ii) Prove P by proving the contrapositive. 3
- (b) Use mathematical induction to prove the following statement for integers $n \geq 1$: 4

$$\sum_{r=1}^n (r-1)^3 \leq \frac{n^4}{4}.$$

- (c) On the complex plane, the line given by $|z+1| = |z-2+3i|$ cuts the circle given by $|z+1| = 3$ into two segments. Find the exact area of the major segment. 4

Question 12 (12 marks) Use a SEPARATE writing booklet.

- (a) For any integer n , prove that $n^2 + n$ is even. 2

- (b) (i) Expand and simplify $(\cos \theta + i \sin \theta)^5$ using the binomial theorem. 2

- (ii) Hence, using de Moivre's theorem, show that 2

$$\sin 5\theta = 16 \sin^5 \theta - 20 \sin^3 \theta + 5 \sin \theta.$$

- (iii) Show that $x = \sin \frac{\pi}{10}$ is one of the solutions to $16x^5 - 20x^3 + 5x - 1 = 0$. 1

- (iv) Write down the polynomial $P(x)$ such that $(x-1)P(x) = 16x^5 - 20x^3 + 5x - 1$. 1

- (v) Find the value of a such that $P(x) = (4x^2 + ax - 1)^2$. 1

- (vi) Hence find the exact value of $\sin \frac{\pi}{10}$. 1

- (vii) Sketch the function $f(x) = 16x^5 - 20x^3 + 5x - 1$, showing all intercepts. 2

Question 13 (12 marks) Use a SEPARATE writing booklet.

(a) $z = 2e^{\frac{\pi}{12}i}$ is a root of the equation $z^4 = a(1 + \sqrt{3}i)$, where a is real.

(i) Express $1 + \sqrt{3}i$ in the form $re^{i\theta}$. 1

(ii) Find the value of a . 1

(iii) Find the other three roots of the equation in the form $re^{i\theta}$. 2

(b) The point P on an Argand diagram represents the complex number z , which satisfies 4

$$\frac{1}{z} + \frac{1}{\bar{z}} = 1.$$

Give a geometrical description of the locus of P as z varies.

(c) For real numbers $a, b > 0$ where $ab \geq 1$, prove that 4

$$\frac{1}{a^2 + 1} + \frac{1}{b^2 + 1} \geq \frac{2}{ab + 1}.$$

Question 14 (12 marks) Use a SEPARATE writing booklet.

(a) If z is one of the fifth roots of unity other than 1, show that 3

$$z - z^2 + z^3 - z^4 = 2i \left(\sin \frac{2\pi}{5} - \sin \frac{\pi}{5} \right).$$

(b) (i) For any integer $a \geq 2$, show that $\frac{1}{a^2} < \frac{1}{a-1} - \frac{1}{a}$. 2

(ii) Hence, for any positive integer n , show that 3

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots + \frac{1}{n^2} < 2.$$

Suggestion: do not use induction.

(c) (i) By considering $f'(x)$ where $f(x) = e^x - x$, show that $e^x > x$ for $x \geq 0$. 1

(ii) Hence, by considering areas under the curves $y = e^x$ and $y = x$,
prove that $\sqrt{e} > \frac{9}{8}$. 1

(iii) Use mathematical induction and the result in (i) to show that, for $x \geq 0$,
 $e^x > \frac{x^n}{n!}$ for all integers $n \geq 1$. 2

End of paper

Q1/ $\sim(p \Rightarrow q) \equiv p \wedge \sim q$
(D)

Q2/ $(4-3i)^2$
 $= 16 - 24i - 9$
 $= 7 - 24i$ (A)

Q3/ $9n^2 - 4 = (3n-2)(3n+2)$

When $n=1$,
 $RHS = 1 \times 5 = 5$ (prime).

For all other n ,
 $RHS = a \times b$ where $a, b > 1$

$\therefore$ no more primes

(B)

Q4/ $|z-2| = 2|z+1|$

$(x-2)^2 + y^2 = 4(x+1)^2 + 4y^2$
 $x^2 - 4x + 4 + y^2 = 4x^2 + 8x + 4 + 4y^2$

$0 = 3x^2 + 12x + 3y^2$

$0 = x^2 + 4x + y^2$

$4 = (x+2)^2 + y^2$

(D)

Q5/

(A)

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 Assessment 2
 Solutions
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Q6/ Let $z = x + iy$ then

$\text{Arg}(z\bar{z}) - \text{Arg}(z - \bar{z})$

$= \text{Arg}(x^2 + y^2) - \text{Arg}(2iy)$

$= 0 - \frac{\pi}{2}$

$= -\frac{\pi}{2}$ (A)

or $\text{Arg } z + \text{Arg } \bar{z} = 0$
 $\text{Arg}(z - \bar{z})$ is $\frac{\pi}{2}$
 using vectors.

Q7/ (B)

Q8/ $(x+iy)^{18} = a + ib$

Now $(y-ix)$

$= \left[i^3 \left(x + \frac{y}{i^3} \right) \right]^{18}$

$= i^{54} \left(x + \frac{y}{-i} \right)^{18}$

$= i^{54} (x + iy)^{18}$

$= (i^4)^{13} \times i^2 \times (a + ib)$

$= 1 \times -1 \times (a + ib)$

$= -a - ib$

(A)

or $(y-ix)^{18} = \left[-i(x+iy) \right]^{18} = i^{18} (x+iy)^{18}$

$= (i^4)^9 \cdot i^2 (x+iy)^{18}$

$= (1)(-1)(x+iy)^{18}$

$= -(x+iy)^{18}$

$= -(a+ib)$

$= -a - ib$

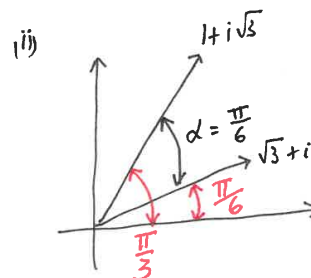
Q9/ $w = 2-3i$
 $\bar{z} = 3+4i$

(a) (i) $\bar{w} + z$
 $= 2+3i + 3+4i$
 $= 5+7i$

(ii) $|w| = \sqrt{4+9}$
 $= \sqrt{13}$

(iii) $\frac{w}{z}$
 $= \frac{2-3i}{3+4i} \times \frac{3-4i}{3-4i}$
 $= \frac{6-8i-9i-12}{9+16}$
 $= \frac{-6-17i}{25}$

(b) (i) $z = (1+\sqrt{3}i) + (\sqrt{3}+i)$
 $= (\sqrt{3}+1) + i(1+\sqrt{3})$



$\theta = \pi - \frac{\pi}{6}$ (co-interior angles are supplementary if parallel lines)
 $\theta = \frac{5\pi}{6}$

(c) Proof by Contradiction.

Suppose that $\log_2 7$ is rational.

Let $\log_2 7 = \frac{p}{q}$ where p and q

are integers.

Then $7 = 2^{\frac{p}{q}}$

so $7^q = 2^p$.

Contradiction!

The LHS is odd while the RHS is even.

$\therefore \log_2 7$ is irrational.

(d) Let the consecutive even powers of 3 be

3^{2n} and 3^{2n+2} , n integer ≥ 1

Then $3^{2n} + 3^{2n+2}$

$= 3^{2n}(1+3^2)$

$= 10 \times 3^{2n}$

$= 10 \times 9 \times 3^{2n-2}$

$= 90 \times 3^{2n-2}$

which is a multiple of 90.
 if n integer ≥ 1

Q10, Ca)

(i) $A = 4\sqrt{2} \operatorname{cis}(\frac{\pi}{4})$

(ii) $B = 4\sqrt{2} \operatorname{cis}(\frac{\pi}{6})$
 $= 4\sqrt{2} (\cos \frac{\pi}{6} + i \sin \frac{\pi}{6})$
 $= 4\sqrt{2} (\frac{\sqrt{3}}{2} + i \frac{1}{2})$
 $= 2\sqrt{6} + 2\sqrt{2}i$

(iii) Now $|5+2i| = \sqrt{29}$
 and $\sqrt{29} < 4\sqrt{2}$ (≈ 5.39)
 it's close enough and a maybe!
 But $\arg(5+2i) = 21^\circ 48' 5''$ so

$(5+2i)$ does not lie in the shaded region because its argument is too small.

(b) (i) Given $\cos \theta + i \sin \theta = e^{i\theta}$
 prove that $(\cos \theta + i \sin \theta)^n$
 $= \cos(n\theta) + i \sin(n\theta)$

Now $(\cos \theta + i \sin \theta)^n$
 $= (e^{i\theta})^n$
 $= e^{in\theta} = e^{i(n\theta)}$
 $= \cos(n\theta) + i \sin(n\theta)$
 as required.

b)(ii) Required to show that

$$(1+i)^n + (1-i)^n = 2(\sqrt{2})^n \cos \frac{n\pi}{4}$$

Now $(1+i)^n + (1-i)^n$
 $= (\sqrt{2} e^{i\frac{\pi}{4}})^n + (\sqrt{2} e^{-i\frac{\pi}{4}})^n$
 $= (\sqrt{2})^n (\cos \frac{n\pi}{4} + i \sin \frac{n\pi}{4}) + (\sqrt{2})^n (\cos \frac{n\pi}{4} - i \sin \frac{n\pi}{4})$
 $= (\sqrt{2})^n \cos \frac{n\pi}{4} + (\sqrt{2})^n \cos \frac{n\pi}{4}$
 $= 2(\sqrt{2})^n \cos \frac{n\pi}{4}$, as required.

(c) All the coefficients are real, so roots occur in conjugate pairs.

(i) The roots are $a+ib, a-ib, a+2ib, a-2ib$ where a, b real

$$\Sigma(x) = -\frac{b}{a} = 4 = 4a \quad \therefore a = 1$$

$$\Pi(x) = (a^2+b^2)(a^2+4b^2) = \frac{c}{a} = 10$$

$$\therefore (1+b^2)(1+4b^2) = 10$$

$$1+5b^2+4b^4 = 10 \quad \text{Let } b^2 = m.$$

$$4m^2+5m-9 = 0$$

$$(4m+9)(m-1) = 0$$

$$m = -\frac{9}{4}$$

$$m = 1$$

$$\therefore b^2 = -\frac{9}{4}$$

$$\therefore b^2 = 1$$

$$\text{But } b \text{ real}$$

$$\therefore b = \pm 1$$

$$a = 1 \text{ and } b = \pm 1$$

(ii) $\therefore P(x) = (x-(1+i))(x-(1-i))(x-(1+2i))(x-(1-2i))$
 $= (x^2-2x+2)(x^2-2x+5)$
 or zeros: $1+i, 1-i$ | $1+2i, 1-2i$
 Sum = 2 | Sum = 2
 Product = 2 | Product = 5

Q11 The statement P is of the form

(i) $(\frac{mn}{2} \text{ is an integer}) \Rightarrow (m \text{ or } n \text{ is even})$

The contrapositive is of the form

$$\sim(m \text{ or } n \text{ is even}) \Rightarrow \sim(\frac{mn}{2} \text{ is an integer}).$$

In words, the contrapositive is:

"If neither m nor n is even, then $\frac{mn}{2}$ is not an integer."

Or: "If m and n are both odd, then $\frac{mn}{2}$ is not an integer."

(ii) Let m and n both be odd.

$$\text{Let } m = 2j+1 \text{ and } n = 2k+1.$$

$$\text{Then } \frac{mn}{2} = \frac{(2j+1)(2k+1)}{2}$$

$$= \frac{4jk+2j+2k+1}{2}$$

$$= 2jk+j+k+\frac{1}{2}$$

Since j and k are integers,

$2jk+j+k$ is an integer.

Hence $2jk+j+k+\frac{1}{2}$ is not an integer.

If both m and n are odd, then $\frac{mn}{2}$ is not an integer.

$\therefore$ If $\frac{mn}{2}$ is an integer, then m or n must be even.

(b) Prove that $\sum_{r=1}^n (r-1)^3 \leq \frac{n^4}{4}$ for integers $n \geq 1$.

when $n=1$, $LHS = \sum_{r=1}^1 (1-1)^3$ and $RHS = \frac{1^4}{4}$
 $= 0$ $= \frac{1}{4}$

Hence $LHS \leq RHS$ when $n=1$.

The statement is true when $n=1$.

Let the statement be true when $n=k$, i.e., let

$$\sum_{r=1}^k (r-1)^3 \leq \frac{k^4}{4}$$

and then try to prove that

$$\sum_{r=1}^{k+1} (r-1)^3 \leq \frac{(k+1)^4}{4}$$

Now $LHS = \sum_{r=1}^{k+1} (r-1)^3$

$$= \sum_{r=1}^k (r-1)^3 + [(k+1)-1]^3$$

$$\leq \frac{k^4}{4} + k^3, \text{ by our assumption}$$

$$= \frac{k^4 + 4k^3}{4}$$

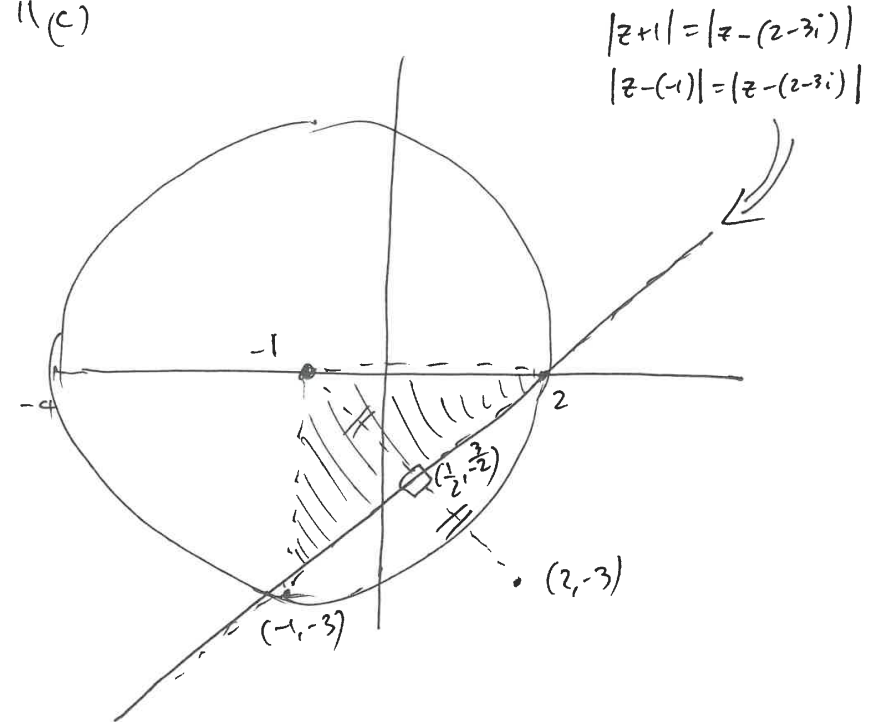
$$\leq \frac{k^4 + 4k^3 + 6k^2 + 4k + 1}{4}$$

$$= \frac{(k+1)^4}{4}$$

Hence $\sum_{r=1}^{k+1} (r-1)^3 \leq \frac{(k+1)^4}{4}$, as required.

Hence the statement is true by mathematical induction.

11(c)



$$|z+1| = |z-(2-3i)|$$

$$|z-(-1)| = |z-(2-3i)|$$

So area of major segment = $\frac{3}{4}$ of circle + shaded triangle
 $= \frac{3}{4} \times \pi \times 2^2 + \frac{1}{2} \times 3 \times 3$
 $= \frac{27\pi}{4} + \frac{9}{2} u^2$

Q17 (a) To prove that n^2+n is even for any integer n .

Case 1 Suppose n is even.

Let $n = 2j$.

$$\text{Then } n^2+n = (2j)^2+2j \\ = 4j^2+2j$$

$$= 2(2j^2+j), \text{ which is even.}$$

Case 2 Suppose n is odd.

Let $n = 2k+1$.

$$\text{Then } n^2+n = (2k+1)^2+(2k+1) \\ = 4k^2+4k+1+2k+1$$

$$= 4k^2+6k+2$$

$$= 2(2k^2+3k+1), \text{ which is even.}$$

$\therefore n^2+n$ is even for any integer n .

$$(b) i) (\cos \theta + i \sin \theta)^5 \\ = \cos^5 \theta + 5 \cos^4 \theta (i \sin \theta) + 10 \cos^3 \theta (i \sin \theta)^2 + 10 \cos^2 \theta (i \sin \theta)^3 + 5 \cos \theta (i \sin \theta)^4 + (i \sin \theta)^5 \\ = \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta + i [5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta]$$

$$(ii) \text{ But } (\cos \theta + i \sin \theta)^5 = \cos 5\theta + i \sin 5\theta$$

$$\text{So } \cos 5\theta = \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta \\ \text{by equating Real parts}$$

$$\text{and } \sin 5\theta = 5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta \\ \text{by equating imaginary parts}$$

$$\therefore \sin 5\theta = 5(1-\sin^2 \theta)^2 \sin \theta - 10(1-\sin^2 \theta) \sin^3 \theta + \sin^5 \theta \\ = 5 \sin \theta - 10 \sin^3 \theta + 5 \sin^5 \theta - 10 \sin^3 \theta + 10 \sin^5 \theta + \sin^5 \theta \\ = 16 \sin^5 \theta - 20 \sin^3 \theta + 5 \sin \theta, \text{ as required.}$$

(iii) The solutions of $16x^5 - 20x^3 + 5x - 1 = 0$

lie amongst the solutions to

$$\sin 5\theta = 1. \text{ if we let } x = \sin \theta$$

$$\text{Solving } \sin 5\theta = 1$$

$$5\theta = \left(\frac{\pi}{2} + 2k\pi\right), k=0,1,2,\dots$$

$$5\theta = \left(\frac{4k\pi + \pi}{2}\right)$$

$$\theta = \frac{4k\pi + \pi}{10}$$

$$\text{When } k=0, \theta = \frac{\pi}{10}.$$

Hence $x = \sin \frac{\pi}{10}$ is one of the solutions of $16x^5 - 20x^3 + 5x - 1 = 0$.

$$(iv) \begin{array}{r} 16x^4 + 16x^3 - 4x^2 - 4x + 1 \\ (x-1) \overline{) 16x^5 + 0x^4 - 20x^3 + 0x^2 + 5x - 1} \\ \underline{16x^5 - 16x^4} \\ 16x^4 - 20x^3 \\ \underline{16x^4 - 16x^3} \\ -4x^3 + 0x^2 \\ \underline{-4x^3 + 4x^2} \\ -4x^2 + 5x \\ \underline{-4x^2 + 4x} \\ x - 1 \\ \underline{x - 1} \\ 0 \end{array}$$

$$\therefore P(x) = 16x^4 + 16x^3 - 4x^2 - 4x + 1$$

(v) $(4x^2 + ax - 1)(4x^2 + ax - 1)$

The coefficient of the term in x^3 is 16.

$$\therefore 4a + 4a = 16$$

$$a = 2$$

(vi) Clearly $\sin \frac{\pi}{10}$ is a solution to $P(x) = 0$

as $\sin \frac{\pi}{10} \neq 1$.

$$\text{Now } P(x) = (4x^2 + 2x - 1)^2$$

$$\therefore \text{Solve } 4x^2 + 2x - 1 = 0$$

$$x = \frac{-2 \pm \sqrt{4 + 16}}{8} = \frac{-2 \pm \sqrt{20}}{8} = \frac{-2 \pm 2\sqrt{5}}{8} = \frac{-1 \pm \sqrt{5}}{4}$$

$$\text{But } \sin \frac{\pi}{10} > 0, \therefore \sin \frac{\pi}{10} = \frac{\sqrt{5} - 1}{4}.$$

(vii) From (iii), other solutions than $\sin \frac{\pi}{10}$ were:

$$k=1: \sin \frac{9\pi}{10} = \sin \frac{\pi}{2} = 1 \text{ which is the } (x-1) \text{ factor}$$

$$k=2: \sin \frac{7\pi}{10} = \sin \frac{\pi}{10} \times \text{duplicate}$$

$$k=3: \sin \frac{5\pi}{10} = -\sin \frac{3\pi}{10} = \frac{-\sqrt{5}-1}{4} \text{ (from (vi))}$$

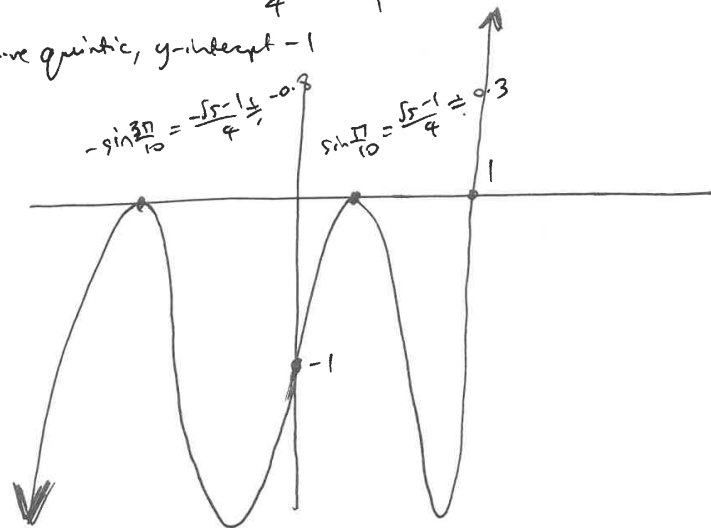
$$k=4: \sin \frac{3\pi}{10} = -\sin \frac{\pi}{10} \times \text{duplicate}$$

$$k=5: \sin \frac{\pi}{10} = \sin \frac{\pi}{10} \times \text{duplicate and around we go again.}$$

From (v) we have 2 double roots.

$$\therefore \text{roots are: } \frac{\sqrt{5}-1}{4}, \frac{\sqrt{5}-1}{4}, \frac{-\sqrt{5}-1}{4}, \frac{-\sqrt{5}-1}{4}, 1$$

Positive quintic, y-intercept -1



Q13 (a)

$$(i) 1 + \sqrt{3}i = 2e^{i\frac{\pi}{3}}$$

$$(ii) \text{ Now } z^4 = a(1 + \sqrt{3}i)$$

$$\therefore (2e^{i\frac{\pi}{3}})^4 = a(2e^{i\frac{\pi}{3}})$$

$$16e^{i\frac{4\pi}{3}} = 2ae^{i\frac{\pi}{3}}$$

$$\therefore a = 8$$

(iii) To find the other 3 roots:

$$z^4 = 8(2e^{i(\frac{\pi}{3} + 2k\pi)})$$

$$z^4 = 16e^{i(\frac{6k\pi + \pi}{12})}$$

$$\text{When } k=-1, \quad z_1 = 2e^{-i\frac{5\pi}{12}}$$

$$\text{When } k=1, \quad z_2 = 2e^{i\frac{7\pi}{12}}$$

$$\text{When } k=2, \quad z_3 = 2e^{i\frac{13\pi}{12}}$$

$$\text{ie, } z_3 = 2e^{-i\frac{11\pi}{12}}$$

(b) Let $z = x + iy$ then

$$\frac{1}{x+iy} + \frac{1}{x-iy} = 1$$

$$\frac{2x}{x^2+y^2} = 1$$

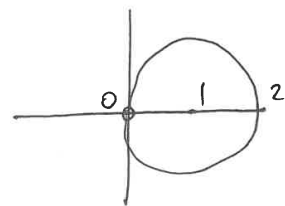
$$2x = x^2 + y^2$$

$$0 = x^2 - 2x + y^2$$

$$0 = (x-1)^2 + y^2 - 1$$

$$1 = (x-1)^2 + y^2$$

The locus of P is a circle, centre (1, 0) and radius = 1.



but a hole at $z=0$ as

$\frac{1}{z}$ & $\frac{1}{z}$ are undefined when $z=0$

(c) Prove that

$$\frac{1}{a^2+1} + \frac{1}{b^2+1} \geq \frac{2}{ab+1} \quad \text{where } a, b > 0 \text{ and } ab \geq 1.$$

Consider LHS - RHS

$$= \frac{1}{a^2+1} + \frac{1}{b^2+1} - \frac{2}{ab+1}$$

$$= \frac{(a^2+b^2+2)(ab+1) - 2(a^2+1)(b^2+1)}{(a^2+1)(b^2+1)(ab+1)}$$

$$= \frac{a^3b + ab^3 + 2ab + a^2 + b^2 + 2 - 2a^2b^2 - 2a^2 - 2b^2 - 2}{(a^2+1)(b^2+1)(ab+1)}$$

$$= \frac{a^3b + ab^3 - a^2 - b^2 + 2ab - 2a^2b^2}{(a^2+1)(b^2+1)(ab+1)}$$

$$= \frac{ab(a^2+b^2) - 1(a^2+b^2) + 2ab(1-ab)}{(a^2+1)(b^2+1)(ab+1)}$$

$$= \frac{(ab-1)(a^2+b^2) - 2ab(ab-1)}{(a^2+1)(b^2+1)(ab+1)}$$

$$= \frac{(ab-1)(a^2-2ab+b^2)}{(a^2+1)(b^2+1)(ab+1)}$$

$$= \frac{(ab-1)(a-b)^2}{(a^2+1)(b^2+1)(ab+1)}$$

which equals 0 when $a=b$,
but is otherwise positive
since $ab \geq 1$.

Hence LHS $\geq$ RHS.

Q14 (a) z is one of the fifth roots
of unity other than 1. So z is

one of

$$\text{cis}\left(\frac{2\pi}{5}\right), \text{cis}\left(\frac{4\pi}{5}\right), \text{cis}\left(\frac{6\pi}{5}\right), \text{cis}\left(\frac{8\pi}{5}\right).$$

$$\text{let } z = \text{cis}\left(\frac{2\pi}{5}\right) \text{ then}$$

$$z - z^2 + z^3 - z^4$$

$$= \text{cis}\left(\frac{2\pi}{5}\right) - \text{cis}\left(\frac{4\pi}{5}\right) + \text{cis}\left(\frac{6\pi}{5}\right) - \text{cis}\left(\frac{8\pi}{5}\right)$$

$$= \cos\left(\frac{2\pi}{5}\right) + i\sin\left(\frac{2\pi}{5}\right) - [\cos\left(\frac{4\pi}{5}\right) + i\sin\left(\frac{4\pi}{5}\right)] + [-\cos\left(\frac{6\pi}{5}\right) - i\sin\left(\frac{6\pi}{5}\right)] - [-\cos\left(\frac{8\pi}{5}\right) + i\sin\left(\frac{8\pi}{5}\right)]$$

$$= \cos\left(\frac{2\pi}{5}\right) + i\sin\left(\frac{2\pi}{5}\right) - \cos\left(\frac{4\pi}{5}\right) - i\sin\left(\frac{4\pi}{5}\right) - \cos\left(\frac{6\pi}{5}\right) - i\sin\left(\frac{6\pi}{5}\right) + \cos\left(\frac{8\pi}{5}\right) - i\sin\left(\frac{8\pi}{5}\right)$$

$$= 2i \sin\left(\frac{2\pi}{5}\right) - 2i \sin\left(\frac{4\pi}{5}\right)$$

$$= 2i \left(\sin\frac{2\pi}{5} - \sin\frac{4\pi}{5} \right), \text{ as required.}$$

(b) (i) Consider RHS - LHS

$$= \frac{1}{a-1} - \frac{1}{a} - \frac{1}{a^2}$$

$$= \frac{a^2 - a(a-1) - (a-1)}{a^2(a-1)}$$

$$= \frac{a^2 - a^2 + a - a + 1}{a^2(a-1)}$$

$$= \frac{1}{a^2(a-1)}, \text{ which is a positive expression as } a \geq 2.$$

Hence RHS $>$ LHS.

$$\therefore \frac{1}{a^2} < \frac{1}{a-1} - \frac{1}{a}$$

$$\begin{aligned}
 \text{(ii)} \quad & \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots + \frac{1}{n^2} \\
 & < \frac{1}{1^2} + \frac{1}{1} - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{1}{n-2} - \frac{1}{n-1} + \frac{1}{n-1} - \frac{1}{n} \\
 & = 2 - \frac{1}{n} \\
 & < 2 \text{ for positive integer } n \\
 \text{Hence } & \frac{1}{1^2} + \frac{1}{2^2} + \dots + \frac{1}{n^2} < 2
 \end{aligned}$$

$$\text{(c) (i)} \quad f(x) = e^x - x$$

$$f'(x) = e^x - 1$$

$$f'(0) = 0$$

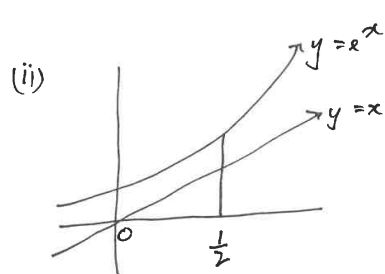
$\therefore (0, 1)$ is a stationary point

$$f''(x) = e^x$$

$$f''(0) = 1 > 0$$

$\therefore (0, 1)$ is a minimum turning point.

$\therefore f(x) = e^x - x$ is increasing for all $x > 0$
 $\therefore$ for $x > 0$, $f(x) > 1$ $\therefore e^x - x > 1$ $\therefore e^x > x + 1$
 $\therefore e^x > x$ for all $x \geq 0$, noting that $e^0 = 1$



noting from (i) that
 $e^x > x$ for $x \geq 0$

$$\text{Now } \int_0^{\frac{1}{2}} e^x dx > \int_0^{\frac{1}{2}} x dx$$

$$[e^x]_0^{\frac{1}{2}} > \left[\frac{x^2}{2}\right]_0^{\frac{1}{2}}$$

$$\sqrt{e} - 1 > \frac{1}{8}$$

$$\therefore \sqrt{e} > \frac{1}{8}$$

$$\therefore \sqrt{e} > \frac{9}{8} \text{ as required.}$$

(iii) To prove that $e^x > \frac{x^n}{n!}$ for all integers $n \geq 1$, $x \geq 0$

$$\text{When } n=1, \text{ LHS} = e^x$$

$$\text{RHS} = \frac{x^1}{1!} = x$$

Since $e^x > x$ from part (i) then $\text{LHS} > \text{RHS}$.

So the statement is true when $n=1$.

Let the statement be true when $n=k$, i.e., let

$$e^x > \frac{x^k}{k!} \text{ for } x \geq 0$$

and then try to prove the statement is true for $n=k+1$, i.e.,

$$\text{try to prove that } e^x > \frac{x^{k+1}}{(k+1)!} \text{ for } x \geq 0$$

$$\text{Given } e^x > \frac{x^k}{k!}$$

$$\text{then } \int_0^t e^x dx > \int_0^t \frac{x^k}{k!} dx$$

$$[e^x]_0^t > \left[\frac{x^{k+1}}{(k+1)!}\right]_0^t$$

$$e^t - 1 > \frac{t^{k+1}}{(k+1)!} - 0$$

$$\therefore e^t > 1 + \frac{t^{k+1}}{(k+1)!}$$

$$\therefore e^t > \frac{t^{k+1}}{(k+1)!}$$

$$\therefore e^x > \frac{x^{k+1}}{(k+1)!} \text{ as required.} \quad \#$$